

## PHYSICS FOR ENGINEER ( 1.1 – 1.21 )

**Ques 1.1:** A spring hang vertically is found to be stretched by 0.02 m from its equilibrium position when a force of 4N acts on it. Then a 2 kg body is attached to the end of the spring and is pulled 0.04m from its equilibrium position along the vertical line. The body is then released and it executes SHM.

- (i) What is the force constant of the spring ?
- (ii) what is the force executed by the spring on the 2kg body just before it is released ?
- (iii) What is the period and frequency of oscillation after release ?
- (iv) what is the amplitude of motion?
- (v) what is the maximum velocity of the oscillating body?
- (vi) What is the mechanical (total) energy of the oscillating system.
- (vii) How much time is required for the body to move half-way into the center from its initial position ?
- (viii) What is the displacement of the body as a function of time ?

**Ques 1.2:** A body is vibrating with SHM of amplitude 15 cm and frequency 4 Hz. Compute (a) the maximum values of the acceleration and velocity and (b) the acceleration and velocity when the displacement is 9 cm

**Ques 1.3:** A particle executes linear harmonic motion about the point  $x = 0$ . At  $t = 0$ , it has displacement  $y = 0.37$  cm and zero velocity. If the frequency of the motion is 0.25/sec. Determine (a) the period (b) the amplitude (c) the maximum speed and (d) the maximum acceleration.

**Ques 1.4:** The displacement of an oscillating particle at an instant  $t$  is given by  $y = a \cos \omega t + b \sin \omega t$  show that it is executing SHM. If  $a = 5$ cm,  $b = 12$ cm and  $\omega = 4$  rad/sec. Calculate (i) the amplitude (ii) the time period (iii) the maximum velocity (iv) the maximum acceleration of the particle

**Ques 1.5:** The position of a particle executing SHM along the  $x$ -axis is  $x = A \cos \omega t$  and  $x = B \sin \omega t$  at time  $t$ . Show that its period of oscillation is given by  $T = (2\pi)/\omega \cos^{-1}(B/2A)$

**Ques 1.6:** For a particle executing SHM the displacement is 8 cm at the instant the velocity is 6 cm/sec and the displacement is 6 cm at the instant the velocity is 8 cm/sec calculate (i) amplitude (ii) frequency (iii) time period

**Ques 1.7:** A SHM is represented by  $y = 10 \sin(10t - \frac{\pi}{6})$  Where  $y$  is measured in meters,  $t$  in seconds and the phase angle in radians. Calculate (i) the frequency (ii) the time period (iii) the maximum displacement (iv) the maximum velocity (v) the maximum acceleration and (vi) displacement, velocity and acceleration at time  $t = 0$  and  $t = 1$  second;

**Ques 1.8:** A particle performs simple harmonic motion given by equation  $y = 20 \sin(\omega t + \alpha)$ . If the time period is 30 seconds and the particle has a displacement of 10 cm at  $t = 0$ ,

find (i) epoch (ii) the phase angle at  $t=5$  seconds and (iii) the phase different between two position of the particle 15 seconds apart.

**Ques 1.9:** A body describing SHM has a maximum accelaeration of  $8\pi \text{ m/s}^2$  and a maximum speed of  $1.6 \text{ m/s}$ . Find the time period  $T$  and the amplitude  $a$ .

**Ques 1.10:** A ball moves in a circular path of  $0.15 \text{ m}$  diameter with a constant angular speed of  $20 \text{ rev/min}$ . Its shadow performs SHM on the wall behind it. Find the acceleration and speed of the shadow (i) at a turning point of the motion (ii) at the equilibrium position and (iii) at a point  $6 \text{ cm}$  from the equilibrium position.

**Ques 1.11:** (i) What is the mechanical (total) energy of the oscillating system of Example 1.1 ?

**Ques 1.12:** A block whose mass is  $680 \text{ gm}$  is at rest on a table top and is fastened to an anchored horizontal spring. The spring constant of the spring is  $65 \text{ N/m}$ . There is negligible friction between the block and the table top. The block pulled a distance  $x = 11 \text{ cm}$  from its equilibrium position at  $x = 0$  and released from rest at  $t=0$

- (i) What force does the spring exert on the block just before the block is released ?
- (ii) What are the angular frequency , the frequency and the period of the resulting oscillation ?
- (iii) What is the amplitude of the oscillation ?
- (iv) What are the maximum velocity and the acceleration of the oscillating block ?
- (v) What is the phase constant  $\Phi$  for the motion ?

**Ques 1.13:** A  $0.2 \text{ kg}$  mass suspended from a spring describes a SHM with a period  $T$  of  $3 \text{ s}$  and amplitude  $R$  of  $10 \text{ cm}$ . At  $t = 0$  the mass passed upward through the equilibrium position. (i) Find the force constant  $k$  of the spring (ii) Find the displacement, velocity and acceleration of the mass when  $t = 1 \text{ s}$ , (iii) Also show that at  $t = 1$ , the sum of the potential and kinetic energy is equal to  $\frac{1}{2}ka$  ?

**Ques 1.14:** A particle performs SHM given by the equation,  $y = 20 \sin(\omega t + \alpha)$  . If the time period is  $30 \text{ sec}$  and the particle has a displacement of  $10 \text{ cm}$  at  $t=0$ . Find (i) epoch (ii) the phase angle at  $t=5$  seconds and (iii) the phase difference between two position of the particle 15 seconds apart.

**Ques 1.15:** A vertically suspended spring of negligible mass and force constant  $R$  is stretched by an amount  $l$  when a body of mass  $m$  is hung on it. The body is pulled by hand an additional distance  $y$  ( positive direction downward) and then released.

- (a) Show that the motion of the body is governed by  $a' = -\frac{k}{m}y$ . So that the body executes harmonic motion about its equilibrium position and
- (b) Show that the period of this motion is the same as that os a simple pendulum of length  $l$ .

**Ques 1.16:** A thin rod of length  $L = 124$  cm and mass  $m = 135$  g is suspended at its midpoint from a long wire. Its period  $T_a$  of angular simple harmonic motion is measured to be  $2.53$  s. An irregular object X is then hung from the same wire and its period  $T_b$  of the angular SHM is found to be  $4.76$  s.

- (i) What is the rotational inertia of object X about its suspension axis ?
- (ii) If both the objects are fastened together and hung from the same wire . What would be the period of oscillation ?

**Ques 1.17:** The scale of a spring balance reading from  $0$  to  $10$  kg is  $0.25$  m long . A body suspended from the balance is found to oscillate vertically with a frequency of  $10/\pi$  hertz. Calculate the mass of the body attached to the spring.

**Ques 1.18:** A body of mass  $0.5$  kg is suspended from a spring of negligible mass and it stretches the spring by  $0.07$  m. For a displacement of  $0.03$  m it has a downward velocity of  $0.4$  m/sec . Calculate (i) the time period (ii) the frequency and (iii) the amplitude of vibration of the spring.

**Ques 1.19:** A simple pendulum of length  $100$  cm has an energy equal to  $2 \times 10^6$  ergs when its amplitude is  $4$  cm. Calculate its energy when (i) its length is doubled and (ii) its amplitude is doubled.

**Ques 1.20:** A small body of mass  $0.10$  kg is undergoing simple harmonic motion of amplitude  $1.0$  metre and period  $0.20$  sec. (i) what is the maximum value of the force acting on it? (ii) if the oscillations are produced by a spring what is the force constant of the spring ?

**Ques 1.21:** An oscillating mass-spring system has a mechanical (total) energy of  $1.0$  joule, amplitude of  $0.10$  metre and maximum speed of  $1.0$  m/sec. Find (i) the force constant of the spring (ii) the mass and (iii) the frequency of oscillation